

A Vertex-Centered Hypergraph-Derived Message Passing Layer Inspired by Eidi-Otter Equivalence Patterns

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Abstract

This note addresses the question of whether one can define a vertex-centered message passing layer whose pairwise interactions are conditioned directly on cardinality-sensitive quantities computed from hyperedge incidence, rather than only on an unweighted clique-expanded graph. Motivated by the structural equivalence framework of Eidi and Otter [4] (in particular their distinction between weak notions, which record only shared neighbors, and stronger notions, which also record the cardinalities of the hyperedges through which neighbors are reached), we propose a vertex-centered, edge-conditioned message passing layer in the sense of Gilmer et al. [5]: it operates on the pairwise neighborhood induced by the hypergraph, but each pairwise interaction is conditioned on two quantities computed directly from hyperedge incidence rather than from a graph projection: the equal-edges random walk weight (a construction that predates Eidi and Otter [4]; see Carletti et al. [1], Zhou et al. [7]) and a cardinality-pattern descriptor introduced here.

The purpose of the construction is to isolate a minimal vertex-centered, hypergraph-derived layer that can be compared conceptually with clique-expansion methods and relational message passing frameworks. In particular, the proposed layer does not start from an influence graph or from a general relational structure, but from the incidence and cardinality data of the hypergraph itself. We show that the layer is permutation equivariant, discuss its computational cost, and identify possible extensions such as curvature-based gates. This provides a first concrete candidate showing how a pairwise MPNN can be enriched with hypergraph-derived, cardinality-sensitive structural information. Further empirical and theoretical validation is needed to assess its expressive and practical advantages.

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1 EO-Inspired Cardinality-Pattern Message Passing

The aim of this short report is to propose a simple vertex-centered, hypergraph-derived message passing layer inspired by the constructions of Eidi and Otter [4]. The guiding principle is that, even when aggregation is ultimately performed over pairwise neighbors, the messages should not depend only on the induced adjacency, but also on a cardinality-sensitive summary of the hyperedges through which two vertices interact.

Standard Message Passing Neural Networks (MPNNs) update node representations by aggregating messages from neighboring nodes [5]. In the graph case, the neighborhood relation is given by edges. In the hypergraph case, however, a hyperedge may contain more than two vertices, and reducing it to pairwise edges may erase part of the higher-order information. This is precisely the distinction that motivates hypergraph neural networks such as AllSet [2], where propagation is performed through node-to-hyperedge and hyperedge-to-node multiset functions.

Our goal here is slightly different. We want a vertex-centered MPNN whose propagation operator is inspired by the equal-edges random-walk viewpoint used by Eidi and Otter [4]. In particular, we use their equal-edges random walk as the basic propagation kernel and augment each message with a structural descriptor encoding the cardinalities of the hyperedges through which the source and target vertices are connected.

This contrasts with the relational-structure viewpoint of Taha et al. [6], where topological message passing is formulated through a finite set of entities, relations, shift operators, and an associated influence graph. That framework is useful as an axiomatic and analytical language, especially for studying oversquashing in topological message passing. However, the construction proposed here does not start from a relational graph or an influence graph. Instead, the message passing rule is defined directly on the hypergraph by using hyperedge incidence and cardinality-sensitive connectivity.

1.1 Hypergraph notation

Let

$$H = (V, \mathcal{E})$$

be a finite undirected hypergraph, where V is the set of vertices and $\mathcal{E} = (e_\alpha)_{\alpha \in A}$ is a finite family of non-empty hyperedges $e_\alpha \in 2^V \setminus \{\emptyset\}$. We allow $\mathcal{E}$ to be a multiset: the same subset of V may occur as several hyperedges, and its multiplicity is recorded (this is relevant for datasets such as co-citation hypergraphs, where one vertex set can be induced by several distinct incidences). For a hyperedge $e \in \mathcal{E}$, its cardinality is denoted by $|e|$. In the implementation, repeated hyperedges are kept as distinct columns of the incidence matrix, whereas each individual hyperedge is treated as a set of distinct vertices.

For a vertex $v \in V$, define the set of incident hyperedges of size at least two by

$$\mathcal{E}(v) := \{e \in \mathcal{E} : v \in e, |e| \geq 2\}.$$

The hypergraph degree of v is

$$d_H(v) := |\mathcal{E}(v)|.$$

For two distinct vertices $v, u \in V$, define the set of hyperedges jointly incident to v and u by

$$\mathcal{E}(v, u) := \{e \in \mathcal{E} : v \in e, u \in e\}.$$

The hypergraph neighborhood of v is then

$$N_H(v) := \{u \in V \setminus \{v\} : \mathcal{E}(v, u) \neq \emptyset\}.$$

Thus, $u \in N_H(v)$ whenever u and v co-occur in at least one hyperedge. However, unlike a clique expansion, we do not forget which hyperedges generate this adjacency. The set $\mathcal{E}(v, u)$ remains part of the message passing rule.

Remark 1.1 (Multiplicities). *The collections $\mathcal{E}(v)$ and $\mathcal{E}(v, u)$ are sub-multisets of $\mathcal{E}$: a subset occurring several times in $\mathcal{E}$ contributes that many elements. Consequently, $d_H(v)$ counts incident hyperedges with multiplicity, and every sum $\sum_{e \in \mathcal{E}(v, u)}$ in the sequel (in Z_{vu} , P^{EE} and χ_{vu}) runs over the family with multiplicity. When $\mathcal{E}$ has no repeated hyperedges, these collections reduce to ordinary sets and all definitions are unchanged.*

1.2 The equal-edges propagation kernel

Eidi and Otter [4] characterize structural and regular equivalences in hypergraphs, distinguishing weak notions (which record only shared neighbors) from stronger notions (which also record the cardinalities of the shared hyperedges). This distinction motivates the two ingredients of the layer proposed here.

The propagation weight used below follows the equal-edges random walk formulation of Coupette et al. [3], which first samples an incident hyperedge and then samples a different vertex uniformly within that hyperedge. This is closely related to earlier two-step hypergraph random walks such as that of Zhou et al. [7], but uses the non-backtracking convention $e \setminus \{v\}$, leading to the factor $|e| - 1$. We follow the equal-edges convention because it underlies the equal-edges Ollivier-Ricci curvature (EE-ORC) in Eidi and Otter [4], the curvature associated with their notion of strong structural equivalence, which motivates the curvature-gate extension discussed in Section 1.9. This random walk defines the probability of moving from a vertex v to another vertex u in one step. Starting at v , the walker first chooses uniformly one hyperedge incident to v . Since there are $d_H(v)$ such hyperedges, each one is chosen with probability $1/d_H(v)$.

If the chosen hyperedge is e , the walker then chooses uniformly one of the $|e| - 1$ vertices of e different from v . Therefore, each hyperedge e containing both v and u contributes a term

$$\frac{1}{d_H(v)} \frac{1}{|e| - 1}$$

to the transition probability from v to u . Summing over all hyperedges containing both vertices gives

$$P^{\text{EE}}(v, u) = \frac{1}{d_H(v)} \sum_{e \in \mathcal{E}(v, u)} \frac{1}{|e| - 1}, \quad u \neq v,$$

whenever $d_H(v) > 0$. If $d_H(v) = 0$ ¹, the neighborhood $N_H(v)$ is empty and the aggregated message $m_v^{(\ell+1)}$ is zero. In the implementation, the update reduces to $h_v^{(\ell+1)} = \phi^{(\ell)}(h_v^{(\ell)}, \mathbf{0})$, which retains $h_v^{(\ell)}$ through ϕ ; equivalently, this corresponds to a self-loop $P^{\text{EE}}(v, v) = 1$ with no incident neighbors.

¹If $d_H(v) = 0$, then v is isolated from the point of view of the equal-edges walk: it is not incident to any hyperedge of size at least two.

Proposition 1.1 (Row-stochasticity of the equal-edges kernel). *Let $v \in V$ be such that $d_H(v) > 0$. Then the equal-edges transition weights define a probability distribution over $N_H(v)$, that is,*

$$\sum_{u \in N_H(v)} P^{\text{EE}}(v, u) = 1.$$

Proof. By definition,

$$P^{\text{EE}}(v, u) = \frac{1}{d_H(v)} \sum_{e \in \mathcal{E}(v, u)} \frac{1}{|e| - 1}.$$

Therefore,

$$\begin{aligned} \sum_{u \in N_H(v)} P^{\text{EE}}(v, u) &= \frac{1}{d_H(v)} \sum_{u \in N_H(v)} \sum_{e \in \mathcal{E}(v, u)} \frac{1}{|e| - 1} \\ &= \frac{1}{d_H(v)} \sum_{e \in \mathcal{E}(v)} \sum_{u \in e \setminus \{v\}} \frac{1}{|e| - 1} \\ &= \frac{1}{d_H(v)} \sum_{e \in \mathcal{E}(v)} 1 = 1. \end{aligned}$$

□

Remark 1.2. *The factor $1/(|e| - 1)$ is essential. It means that a neighbor reached through a small hyperedge receives more weight than a neighbor reached through a very large hyperedge. Therefore, P^{EE} is not simply the random walk on the clique expansion. It is a cardinality-sensitive hypergraph propagation operator.*

1.3 Cardinality-pattern summary

The transition kernel $P^{\text{EE}}(v, u)$ determines how much weight vertex v assigns to the neighbor u during aggregation. However, this scalar weight does not fully describe the way in which v and u are connected. As emphasized by Eidi and Otter [4], hypergraph connectivity can be considered at different levels of resolution: weak notions remember only the existence of shared neighbors, whereas stronger notions also take into account the cardinalities of the hyperedges through which those neighbors are reached.

Motivated by this distinction, we augment the equal-edges transition weight with a fixed, structural cardinality summary. Two vertices may be connected through a small hyperedge, through a large hyperedge, or through several hyperedges of different cardinalities. These cases should not necessarily produce the same type of message, even when their aggregated equal-edges weights are similar.

To define this summary, fix an integer $K \geq 2$, corresponding in the implementation to `max_card`. We use $K - 1$ cardinality bins: the first bins represent hyperedge sizes $2, 3, \dots, K - 1$, while the last bin represents all sizes at least K . For each cardinality $r \geq 2$, let $\mathbf{c}_r \in \{0, 1\}^{K-1}$ be the corresponding one-hot cardinality encoding. Its nonzero entry is placed at position

$$b(r) = \min(r, K) - 1,$$

where positions are indexed from 1. This encoding is fixed and parameter-free.

For each pair v, u with $\mathcal{E}(v, u) \neq \emptyset$, define the normalizing factor

$$Z_{vu} := \sum_{e \in \mathcal{E}(v, u)} \frac{1}{|e| - 1}.$$

We then define the cardinality-pattern descriptor

$$\chi_{vu} := \frac{1}{Z_{vu}} \sum_{e \in \mathcal{E}(v, u)} \frac{1}{|e| - 1} \mathbf{c}_{|e|}.$$

Thus, χ_{vu} is a fixed, $1/(|e| - 1)$ -weighted histogram over the cardinality bins of the hyperedges shared by v and u . It carries no learnable parameters of its own: it is a deterministic function of the hypergraph structure. All learning is delegated to the message function ψ , which receives χ_{vu} as a structural input and maps it to a useful representation.

The role of χ_{vu} is therefore to refine pairwise adjacency with cardinality information from the original hypergraph. It does not encode the full multiset of shared hyperedges, and different local incidence patterns may lead to the same descriptor. Rather, it provides a cardinality-sensitive summary that allows the message function to distinguish some hypergraphs that would be identical after an unweighted clique expansion.

Example 1.1. *For example, consider two hypergraphs on vertices $\{a, b, c\}$:*

$$H_1 : \mathcal{E}_1 = \{\{a, b, c\}\}, \quad H_2 : \mathcal{E}_2 = \{\{a, b\}, \{a, c\}, \{b, c\}\}.$$

Both induce the same clique expansion, namely a triangle. However, for the pair (a, b) , the first hypergraph gives a cardinality pattern corresponding to a hyperedge of size 3, while the second gives a cardinality pattern corresponding to a hyperedge of size 2. Hence

$$\chi_{ab}^{H_1} = \mathbf{c}_3, \quad \chi_{ab}^{H_2} = \mathbf{c}_2.$$

Thus, the proposed layer can distinguish these two hypergraphs despite their identical unweighted clique projection. This illustrates that χ_{vu} acts as a cardinality-sensitive refinement of pairwise adjacency, although not as a complete invariant of higher-order incidence structure.

We use a fixed one-hot cardinality encoding rather than a separate learnable cardinality embedding. This keeps the descriptor parameter-free and makes the structural information exposed to the model explicit. Since concatenating χ_{vu} increases the input dimension of the message MLP, improvements over variants without χ_{vu} cannot by themselves be attributed solely to meaningful hypergraph structure. For this reason, the experiments include a shuffled- χ control, which preserves the same input dimension while destroying the meaningful assignment of cardinality patterns to vertex pairs.

1.4 The EO-Pattern MPNN layer

Each vertex $v \in V$ carries a hidden representation

$$h_v^{(\ell)} \in \mathbb{R}^{d_\ell}$$

at layer ℓ . The goal of one message passing layer is to update this representation by aggregating information from the hypergraph neighbors of v .

For each neighbor $u \in N_H(v)$, we construct a message using three pieces of information: the current representation of the target vertex v , the current representation of the source vertex u , and the cardinality-sensitive descriptor χ_{vu} summarizing the sizes of the hyperedges through which v and u are jointly incident. This gives a learnable message of the form

$$\psi^{(\ell)}\left(h_v^{(\ell)}, h_u^{(\ell)}, \chi_{vu}\right),$$

where $\psi^{(\ell)}$ may be implemented, for instance, as an MLP.

Definition 1.1 (EO-Pattern MPNN layer). *The EO-Pattern MPNN layer maps hidden representations $\{h_v^{(\ell)}\}_{v \in V}$ to $\{h_v^{(\ell+1)}\}_{v \in V}$ via the following update rule for each $v \in V$:*

$$m_v^{(\ell+1)} = \sum_{u \in N_H(v)} P^{\text{EE}}(v, u) \psi^{(\ell)}\left(h_v^{(\ell)}, h_u^{(\ell)}, \chi_{vu}\right),$$

$$h_v^{(\ell+1)} = \phi^{(\ell)}\left(h_v^{(\ell)}, m_v^{(\ell+1)}\right),$$

where $\psi^{(\ell)}$ and $\phi^{(\ell)}$ are learnable functions of compatible dimensions.

A simple concrete choice is

$$h_v^{(\ell+1)} = \text{MLP}^{(\ell)}\left(h_v^{(\ell)} \parallel m_v^{(\ell+1)}\right),$$

where $\parallel$ denotes concatenation.

This architecture has the standard message/aggregate/update structure of a Gilmer et al. [5] MPNN. However, the interaction between v and u is not determined only by their adjacency in the induced pairwise graph. The scalar weight $P^{\text{EE}}(v, u)$ is obtained from the equal-edges hypergraph random walk, while the message function is conditioned on the cardinality-pattern descriptor χ_{vu} . Hence, the layer uses the induced neighborhood as the support of aggregation, but the propagation rule itself remains sensitive to the hypergraph structure that generated that neighborhood.

1.5 In what sense this is hypergraph-derived

The proposed layer is an edge-conditioned MPNN. The aggregation support is the clique-expanded pairwise neighborhood $N_H(v)$; what goes beyond a standard clique-expansion GNN is that the edge weights and edge descriptors are computed from the original hypergraph incidence structure. Its propagation rule uses two quantities defined intrinsically from the hypergraph:

1. The equal-edges transition weight $P^{\text{EE}}(v, u)$.
2. The cardinality-pattern descriptor χ_{vu} .

The first quantity controls the strength of propagation from u to v . The second controls the structural type of the message. Together, they encode not only that two vertices are adjacent through the hypergraph, but also a cardinality-sensitive summary of the incident hyperedges that generate this adjacency. Therefore, the construction is hypergraph-derived in a precise but limited sense: the support of aggregation is the same as in a clique-expansion MPNN, but the edge weights and message descriptors are computed directly from the original hypergraph incidence structure rather than from an unweighted graph projection.

This is the key difference with clique expansion. A clique expansion keeps only the fact that two vertices co-occur in at least one hyperedge. The proposed layer also keeps information about the cardinalities of those hyperedges. Therefore, it is sensitive to selected cardinality information that an unweighted pairwise graph projection may erase.

1.6 Relation to relational message passing

The construction should also be distinguished from the relational-structure framework of Taha et al. [6]. In that framework, a higher-order object is represented as a relational structure

$$R = (S, R_1, \dots, R_k),$$

with shift operators associated with the relations. These shift operators are aggregated into influence matrices, which in turn define an influence graph: a directed weighted graph on the entity set S , with an edge $\tau \rightarrow \sigma$ whenever the aggregated influence of τ on σ is non-zero. This influence graph is then used to analyze information propagation and oversquashing.

The present construction does not use a relational structure as its starting point. It starts directly from the hypergraph incidence pattern. The message passing rule is determined by $\mathcal{E}(v, u)$, $|e|$, P^{EE} , and χ_{vu} . Of course, as with many finite message passing schemes, one could later encode this model inside a sufficiently rich relational framework. But this would be an after-the-fact representation, not the conceptual origin of the architecture.

Thus, the distinction is not that the proposed layer is impossible to relationalize. The distinction is that the primitive quantities defining propagation are computed directly from hypergraph incidence and hyperedge cardinalities.

1.7 Permutation equivariance

Proposition 1.2 (Permutation equivariance). *The proposed layer is permutation equivariant with respect to relabelings of the vertex set.*

Proof. Let $\pi : V \rightarrow V$ be a permutation of the vertices, and let πH be the relabeled hypergraph with hyperedges

$$\pi \mathcal{E} := \{\pi(e) : e \in \mathcal{E}\}, \quad \pi(e) := \{\pi(v) : v \in e\}.$$

Then

$$\mathcal{E}_{\pi H}(\pi v, \pi u) = \{\pi(e) : e \in \mathcal{E}_H(v, u)\}.$$

Moreover, $|\pi(e)| = |e|$ and $d_{\pi H}(\pi v) = d_H(v)$. Hence

$$P_{\pi H}^{\text{EE}}(\pi v, \pi u) = P_H^{\text{EE}}(v, u),$$

and

$$\chi_{\pi v, \pi u}^{\pi H} = \chi_{vu}^H.$$

Assume also that the hidden states are relabeled accordingly, so that

$$h_{\pi v}^{(\ell)}(\pi H) = h_v^{(\ell)}(H) \quad \text{for all } v \in V.$$

Since the message function $\psi^{(\ell)}$ and the update function $\phi^{(\ell)}$ are shared across all vertices, and since aggregation is performed by summation over the unordered neighborhood of each vertex, the resulting layer satisfies

$$h_{\pi v}^{(\ell+1)}(\pi H) = h_v^{(\ell+1)}(H).$$

Therefore, the layer is vertex permutation equivariant. $\square$

1.8 Computational cost

Proposition 1.3 (Computational cost). *Let*

$$S_H := \sum_{e \in \mathcal{E}} |e|(|e| - 1)$$

be the number of ordered vertex pairs generated by the hyperedges, and let

$$M_H := \sum_{v \in V} |N_H(v)|$$

be the number of directed edges in the clique-expanded support. Then the quantities $P^{\text{EE}}(v, u)$ and χ_{vu} can be precomputed in time $O(S_H)$, up to the cost of storing the $(K - 1)$ -dimensional descriptors. Once these quantities are precomputed, one forward pass of the proposed layer has the same aggregation support as a clique-expansion MPNN, namely M_H directed pairwise interactions.

Proof. For each hyperedge $e \in \mathcal{E}$, the construction updates the quantities associated with all ordered pairs (v, u) such that $v, u \in e$ and $v \neq u$. The number of such ordered pairs is $|e|(|e| - 1)$. Summing over all hyperedges gives a total preprocessing cost proportional to

$$\sum_{e \in \mathcal{E}} |e|(|e| - 1) = S_H.$$

These updates accumulate the scalar contributions defining $P^{\text{EE}}(v, u)$ and the corresponding cardinality-bin contributions defining χ_{vu} .

After this preprocessing step, the layer aggregates messages only over pairs (v, u) with $u \in N_H(v)$. Therefore the number of message evaluations in one forward pass is

$$\sum_{v \in V} |N_H(v)| = M_H,$$

which is exactly the number of directed edges in the clique-expanded support. The additional hypergraph information appears in the precomputed edge weights and edge descriptors, not in a larger aggregation support. $\square$

1.9 Possible extensions

The base layer deliberately avoids additional mechanisms such as hyperedge hidden states, hyperedge-to-hyperedge propagation, quotient-level summaries, or curvature gates. These extensions may be useful, but they are not necessary to define the core vertex-centered hypergraph-derived layer.

A natural extension inspired by Eidi and Otter [4] would be to include an Ollivier-Ricci curvature gate. If $\kappa_{\text{EE}}(v, u)$ denotes the equal-edges Ollivier-Ricci curvature associated with the pair (v, u) , one could replace the propagation weight by

$$\tilde{P}_{\text{EE}}(v, u) = P^{\text{EE}}(v, u) g(\kappa_{\text{EE}}(v, u)),$$

where g is either a fixed scalar function or a learnable gate. This would make the layer sensitive not only to local hyperedge cardinalities, but also to the local geometry induced by optimal transport. Nevertheless, this curvature-gated version should be considered an extension, not the minimal model. The goal of this first proposal is to isolate the minimal vertex-centered hypergraph-derived message passing layer and to check whether it is mathematically well-defined and implementable. More sophisticated mechanisms can then be studied once the core model has been validated, both theoretically and empirically.

1.10 Summary

The proposed EO-Pattern MPNN is the layer

$$m_v^{(\ell+1)} = \sum_{u \in N_H(v)} P^{\text{EE}}(v, u) \psi^{(\ell)}(h_v^{(\ell)}, h_u^{(\ell)}, \chi_{vu}),$$

$$h_v^{(\ell+1)} = \phi^{(\ell)}(h_v^{(\ell)}, m_v^{(\ell+1)}),$$

where

$$P^{\text{EE}}(v, u) = \frac{1}{d_H(v)} \sum_{e \in \mathcal{E}(v, u)} \frac{1}{|e| - 1},$$

and

$$\chi_{vu} = \frac{1}{Z_{vu}} \sum_{e \in \mathcal{E}(v, u)} \frac{1}{|e| - 1} \mathbf{c}_{|e|}, \quad Z_{vu} = \sum_{e \in \mathcal{E}(v, u)} \frac{1}{|e| - 1}.$$

This gives a simple, vertex-centered, permutation-equivariant message passing layer. Its main feature is that messages are weighted by equal-edges transition weights and conditioned on the cardinality pattern of the hyperedges through which vertices communicate. In this precise sense, the model uses hypergraph-derived incidence and cardinality information rather than only an unweighted clique projection. More precisely, it remains a vertex-centered pairwise aggregation scheme, but the pairwise weights and message descriptors are induced by the original hyperedge incidence and cardinality structure.

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